Roll No __________ Subject Code: SCR3S3

Institute of Engineering & Technology,
Devi Ahilya University, Indore
Main Exam Dec 2023
B.E. (CS) 2 nd Year
Exam Code 2024025
Environment Studies - E-WASTES MANAGEMENT

Max marks: 60

Time: 3 Hrs

NOTE:

1) Attempt all 5 questions. All the questions carry equal marks.
2) Draw double under at the end of each question.
3) Mention question No. Clearly as per question paper.
4) Whichever questions you are going to attempt that must be properly marked/checked on top of the answer sheet.

Q1) What is & how is Electronic waste generated.

OR

Q2) Cisco Inc. , Founders Leonard Bosack & Sandy Lerner, in San Francisco, California, United States appointed you to prepare a Business Model for E-Waste. Do the same in the answer sheet.

Q3) What is the basis for defining E-waste, & what are the possible hazardous substances present in E-waste?

OR

Q4) Explain RoHS in the Electronic & electrical Equipments & also explain EPR.

Q5) What is SWOT analysis? Explain.

OR

Q6) Explain Basel Convention European Union policy and regulation on e-waste US Policy for E-Waste.

Q7) While making and implementing new legislation and initiate the use of suitable instruments for e-waste management 8 goals have been stated, what are those 8 goals, write only name & explain the scope of products included in the currently operating WEEE management systems.

OR

Q8) Do SWOT analysis for Best Artificial Intelligence Companies in India like 1) Tata Elxsi Ltd. 2) Bosch Ltd. 3) Kellton Tech Solutions Ltd. 4) Happiest Minds Technologies Ltd. 5) Zensar Technologies Ltd. whether these companies can play an important role at global level use the following other information for your case -

The India artificial intelligence market size reached US$ 680.1 Million in 2022. Looking forward, IMARC Group expects the market to reach US$ 3,935.5 Million by 2028, exhibiting a growth rate (CAGR) of 33.28% during 2023-2028. The growing pool of skilled talent in STEM fields, the introduction of numerous favorable government initiatives, rapid digitization, the increasing demand for AI-driven solutions across various industrial verticals, and continuous innovations are among the key factors driving the market growth.

Artificial Intelligence, or AI, refers to a subdivision of computer science that is focused on the creation of intelligent machines apt to perform tasks that typically require human intelligence. It entails the development of algorithms and models that enable machines to learn from data, reason, plan, and adapt to new situations. AI technologies encompass various subfields, including machine learning (ML), natural language processing (NLP), computer vision, and robotics. AI is widely applied in diverse sectors, such as automation, healthcare, finance, and transportation, revolutionizing industries by expanding human capabilities and enabling machines to execute complex tasks efficiently and autonomously.

The AI market in India is being driven by the vast pool of skilled talent in science, technology, engineering, and mathematics (STEM) fields that contributes to the creation of innovative AI algorithms, applications, and solutions that cater to both domestic and global markets. Moreover, the government's proactive approach to promote AI adoption through policy initiatives and funding support is bolstering the market expansion. In line with this, the introduction of the National AI Strategy and Startup India have encouraged investments in AI startups and facilitated collaborations between academia and industry, thus contributing to the market growth. Additionally, the rapid digital transformation across industries, including finance, healthcare, e-commerce, and manufacturing, has spurred the demand for AI-driven solutions to enhance efficiency, productivity, and customer experience, which, in turn, is strengthening the market growth.

India possesses a strong pool of skilled professionals in the STEM stream, owing to its education system that emphasizes technical education. This has resulted in numerous graduates with expertise in computer science, data science, and related fields, creating a solid foundation for AI research and development. India's skilled talent actively engages in cutting-edge AI research, leading to the development of tailored AI solutions for specific industries and challenges. The abundance of talent has also fueled the growth of AI startups, supported by incubators and venture capital firms. In addition to this, India's skilled AI workforce is sought after globally, leading to the establishment of offshore AI development centers, enhancing the country's position in the global AI market.

The Indian government acknowledges the transformative potential of AI and has taken proactive measures to encourage its adoption across industries. They unveiled a National AI Strategy, outlining a roadmap for AI development, talent nurturing, and robust infrastructure building. This commitment instills confidence among businesses and investors, leading to increased interest in AI projects and aiding in market expansion. The government's 'Startup India' program supports entrepreneurship and innovation by providing financial assistance, tax benefits, and simplified regulatory processes to AI-driven startups, further strengthening the market growth. Concurrent with this, the government is also fostering collaborations between academia and industry, facilitating technology transfer and knowledge exchange, and expediting AI applications across sectors. These initiatives collectively boost India's position in the global AI landscape.

Q9) Explain the case study of Bridgestone Tyre that was discussed in the classroom.

OR

Q10) What are markets for recycling technology?